

5. (a) Discuss the role of ATP and calcium ions in the excitation-contraction coupling in a skeletal muscle fibre with help of diagrams. (10)
- (b) Give a brief account of the physiological effect of the hormones secreted by the thyroid gland. (5)
6. Write short notes on any **three** of the following : (3×5=15)
- (a) Oogenesis
- (b) Sliding Filament mechanism
- (c) Hypothalamo-hypophyseal portal system
- (d) Implantation
- (e) Neuromuscular junction

(1000)

[This question paper contains 4 printed pages.]

07.06.2024(M)
Your Roll No.....

Sr. No. of Question Paper : 4164

Unique Paper Code : 2232011203

Name of the Paper : Human Physiology, Control and Coordination Systems/ Discipline Specific Core- 6

Name of the Course : B.Sc (Hons) Zoology Exam- 2023

Semester : II – (NEP-UGCF)

Duration : 2 Hours

Maximum Marks : 60

Instructions for Candidates

- Write your Roll No. on the top immediately on receipt of this question paper.
- Attempt any **FOUR** QUESTIONS in all and Question No. 1 is **COMPULSORY**.

- (a) Define the following terms (**Any four**) : (4)

(i) Sertoli cells

(ii) Hyperpolarization

P.T.O.

(iii) Sarcomere

(iv) Trigger zone

(v) Perimysium

(b) Differentiate between the following : (4)

(i) Myelinated vs Unmyelinated nerve fibre

(ii) Follicular phase vs Luteal phase

(c) Give the location and function of the following : (3)

(i) Islet of Langerhans

(ii) Sarcoplasmic reticulum

(iii) Nissl granules

(d) Expand the following : (2)

(i) PNS

(ii) FSH

(iii) EPSP

(iv) Ach

(e) Fill in the blanks : (2)

(i) _____ cells form the myelin sheath around the neurons of the PNS.

(ii) The mature and dominant follicle in the ovary is known as _____.

2. (a) What is action potential? Explain the generation of action potential in a neuron. (10)

(b) Explain the all or none principle of action potential. (5)

3. (a) What is Spermatogenesis? Explain the process of spermatogenesis. (10)

(b) Draw a well-labelled diagram of the mature mammalian sperm. (5)

4. (a) How does pancreas regulate the normal level of glucose in the blood? (10)

(b) Describe the negative feedback regulation of secretion of a hormone. (5)